

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1102

COURSE OUTCOME COVERED

CO4: Implement object-oriented system layers using design principles and patterns, and integrate access and view layers with OODBMS (*BL3*)
Assessment Tool Weightage: 30%

QUESTIONS: 15

Total Marks:30

Annexure A

CLASS TEST

Code of Conduct:

I Sayed Fazal (192311291) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

CLASS TEST

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Annexure A

CLASS TEST

QUESTIONS:

Total Marks: 30

Q1. Define architectural patterns in object-oriented systems. List any two common patterns. Explain how the Model-View-Controller (MVC) architecture improves modularity and maintainability.

[L1, L2, L3]

Q2. What is the Data Access Object (DAO) Pattern? Mention two advantages. Explain how DAO separates database-related operations from application logic.

[L2, L3, L4]

Q3. What are object-oriented design guidelines? Identify any two categories of design guidelines. Explain the role of the Liskov Substitution Principle (LSP) in developing reusable software components.

[L2, L3, L4]

Q4. Define database persistence in object-oriented applications. Mention two persistence mechanisms. Explain how an Object-Relational Mapping (ORM) framework supports persistent object management.

[L1, L2, L3]

Q5. What is an Application Layer? State two major responsibilities. Explain how the Application Layer coordinates use cases without directly implementing domain business rules.

[L2, L3, L4]

Q6. Explain the structure of a three-tier architecture consisting of Presentation, Business, and Data layers. Analyze the flow of a customer transaction through these layers in an e-commerce application.

[L2, L4, L5]

Q7. What is the role of a Business Logic Layer? List two important functions. Explain how it validates and processes requests received from the presentation layer.

[L2, L3, L4]

Q8. Explain the significance of cohesion and coupling in object-oriented design. Analyze how high cohesion and low coupling contribute to software maintainability and reusability.

[L3, L4, L5]

Q9.What is encapsulation in an object-oriented system? Explain how encapsulation protects object state and compare it with data hiding using a suitable example.

[L2, L4]

Q10.Design a layered object-oriented system for an online banking application. Identify the major layers, recommend suitable design patterns, and justify how the architecture improves security, scalability, and maintainability.

[L3, L4, L5]

Q11.What is the role of the Infrastructure Layer in object-oriented architecture? List two responsibilities. Explain how it supports technical services such as database access and external system communication.

[L1, L2, L3]

Q12.Define the Singleton Design Pattern. Mention its major characteristics. Explain its appropriate use in managing shared resources and discuss one limitation of the pattern.

[L2, L3, L4]

Q13.What is dependency injection in object-oriented design? List two benefits. Analyze how dependency injection reduces tight coupling between classes and improves testability.

[L2, L3, L4]

Q14.Explain the working of the Strategy Design Pattern. Compare the roles of Context and Strategy with a suitable real-world application example.

[L3, L4, L5]

Q15.Design a layered architecture for a Library Management System. Identify suitable layers, recommend relevant design patterns, and justify how your proposed architecture supports flexibility and maintainability.

[L3, L4, L5]

1A) Data Access Object (DAO)

The data access object (DAO) pattern is a structural design pattern that provides an abstract interface for accessing and manipulating data in a database.

Advantages of DAO

1. Separation of Concerns Database operations are kept separate from business / application logic, making the code easier to understand and maintain.
2. Easy database changes The database implementation can be changed without making major changes to the applications logic.

How DAO separates database operations from application logic:

The DAO pattern uses a separate DAO class / interface to handle all database operations. The application layer only calls DAO methods and does not directly interact with SQL queries or database connections.

Example

Application Logic



User Service → UserDao → Database

2A) Application Layer

The application layer is the layer in a software system that coordinates and manages application use cases. It acts as a bridge between the user/interface layer and the domain layer.

Two major responsibilities

1. Coordinate use case - It controls the sequence of operations required to complete a specific use case, such as placing an order or registering a customer.
2. Manage application-level - It handles tasks of application layer such as receiving input, calling domain objects, managing transactions and returning results.

⇒ The Application layer does not contain the core business rules. Instead it delegates those rules to domain entities and domain services.

Example

User Interface



Application layer



Order Service



DAO / Database

3A) Role of Business Logic layer

The Business Logic layer (BL1) is the part of an application that contains the business rules, policies and processing logic of the system. It acts as an intermediate layer between the presentation layer and the Data Access layer.

Two Important functions

1. Validation - It checks whether the data received from the presentation layer is valid and follows the business rules.
2. Processing - It performs the required business operations, calculations, decisions and workflow before storing or retrieving data.

How does it validate and process requests?

⇒ When a user sends a request through the presentation layer, the request is passed to the Business layer. The BL1 first validates the input according to business rules. If the input is valid, it processes the request.

Example

Presentation layer



Business Logic layer



Validation



Processing

→ Data Access layer → Database

4A) Encapsulation

Encapsulation is an object-oriented programming concept of combining data (attribute) and the methods that operate on the data into single unit called a class. It also controls how the internal state of an object can be accessed or modified.

Encapsulation vs. Data Hiding

Encapsulation

1) Combines data and methods into a single class

2) Focuses on bundling and controlled access

3) Achieved using classes, methods and access modifiers

Data Hiding

1) Restricts direct access to internal data.

2) Focuses mainly on protecting implementation

3) Mainly achieved using private, protected etc.

Example

```
class BankAccount {
```

```
    private double balance;
```

```
    public void deposit (double amount) {
```

```
        if (amount > 0) {
```

```
            balance += amount;
```

```
        }  
    }
```



```
public double getBalance () {
```

```
    return balance;
```

```
}
```

```
}
```

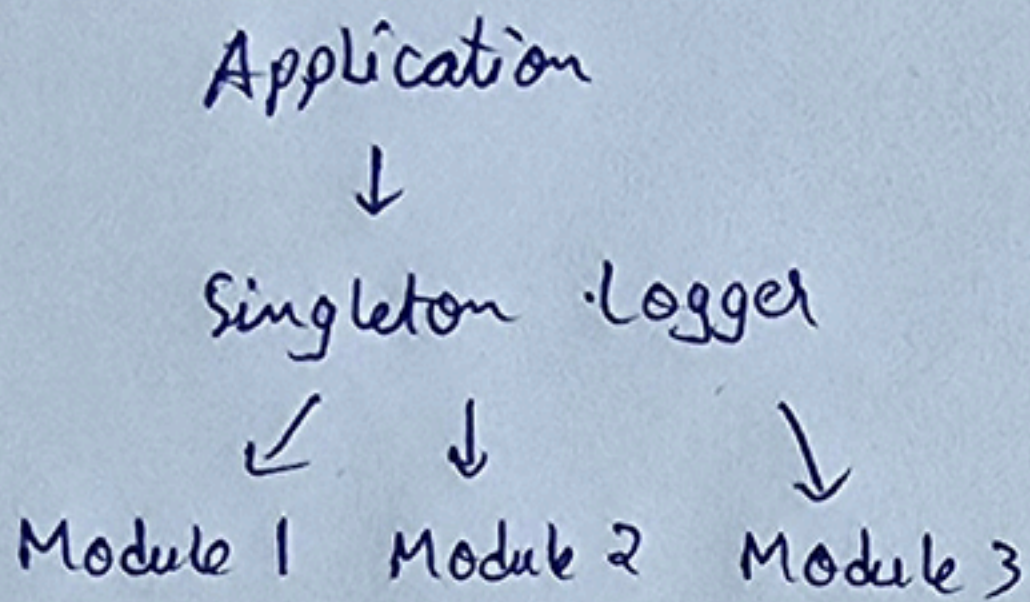
5A) Singleton Design Pattern

The Singleton Design pattern is a creational design pattern that ensures a class has only one instance throughout the application and provides a global access point to that instance.

Major characteristics

- 1) Single Instance - Only one object of the class can be created.
- 2) Global Access - The instance can be accessed from anywhere in the application.
- 3) Private Constructor - The constructor is usually private to prevent direct object creation.
- 4) Static instance - A static variable is used to store the single instance.

Example



One Limitation

A major limitation is that Singleton introduces global state, which can make the system harder to test and maintain. Different parts of the application become dependent on the same shared instance, increasing coupling.